



ORION 1.0

Problem Statement ID: ORION-PS-03

Problem Statement Title: SylvaSense

Team ID:

Team Name: Order of One



Proposed Solution

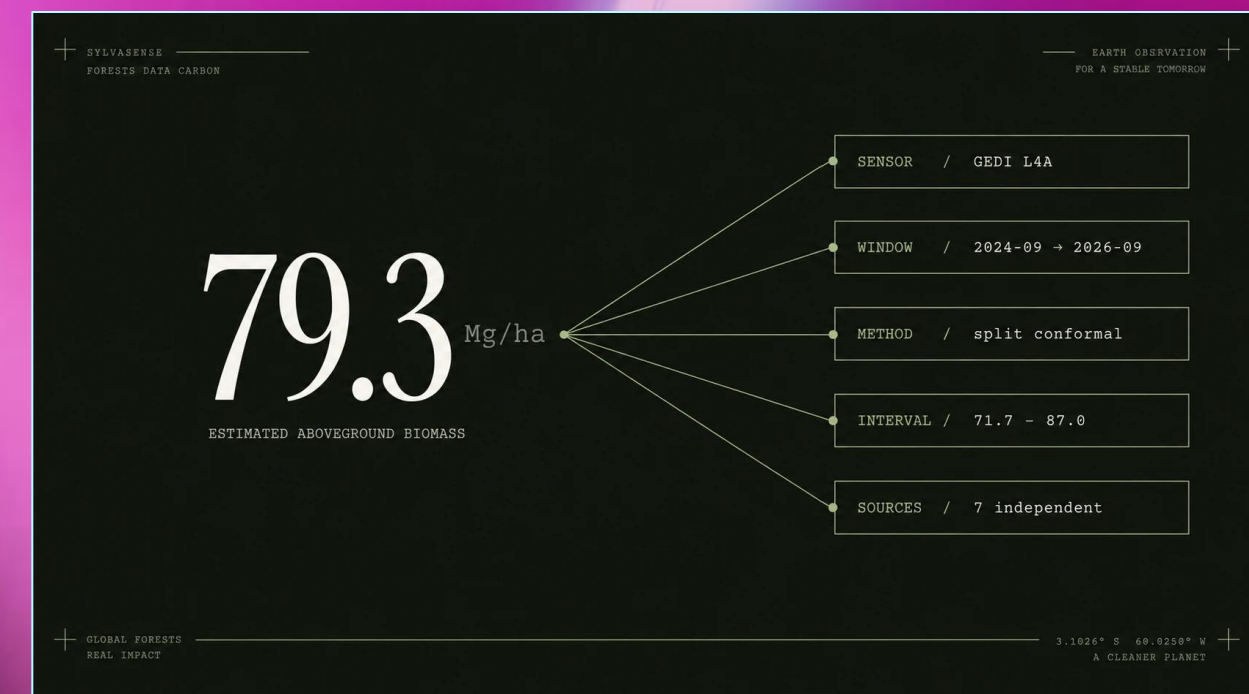
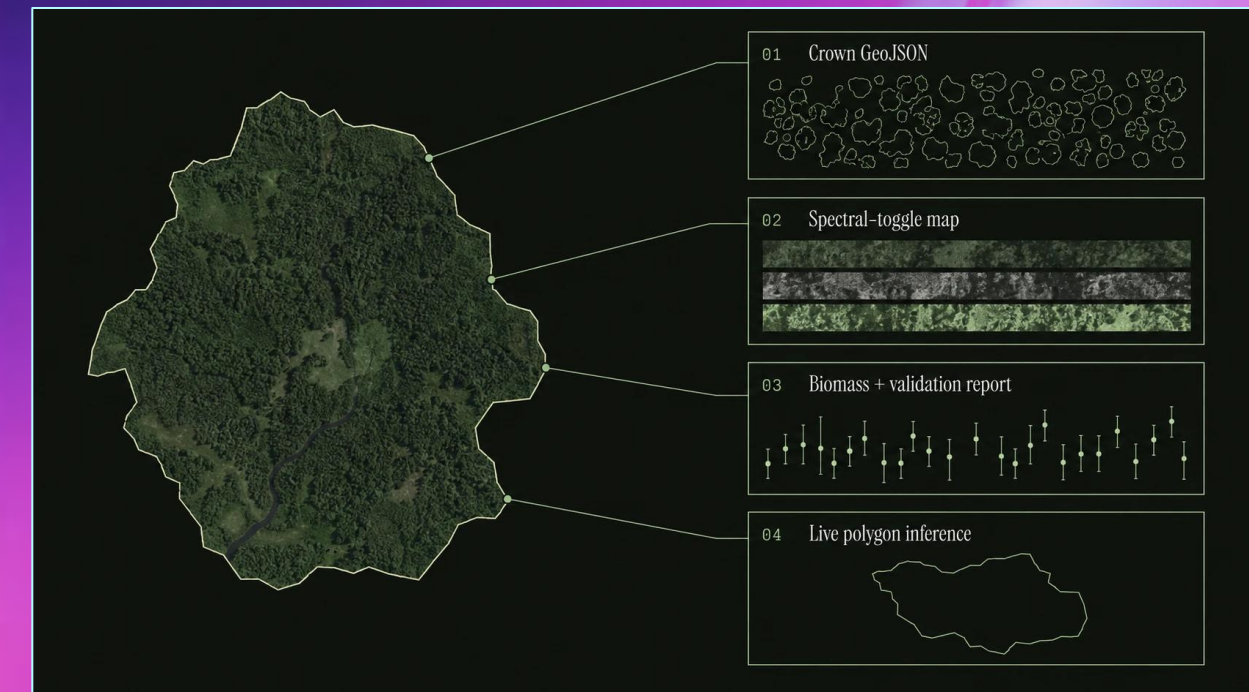
SYLVASENSE

From canopy to carbon.

One polygon in. Four kinds of evidence out.
Every number carries its uncertainty.

- 01 Crown GeoJSON**
Per-tree polygons, each with a confidence score
- 02 Spectral-toggle map**
Optical, radar and structure in one frame
- 03 Biomass + validation report**
Mg/ha with an interval that is checked
- 04 Live polygon inference**
Draw an area; evidence returns with its sources

[Live demo → sylvasense-orion.vercel.app](https://sylvasense-orion.vercel.app)





Technical Approach

Three scales. One pipeline.

CROWNS / FINE AERIAL IMAGERY

Mask2Former + Swin-S

0.1 m NEON RGB, tiled and merged

Every mask checked against the canopy height model

BIOMASS / OPTICAL + SAR + LIDAR

Gradient boosting + split conformal

GEDI footprints are the calibration target

Coverage measured on held-out ground, not assumed

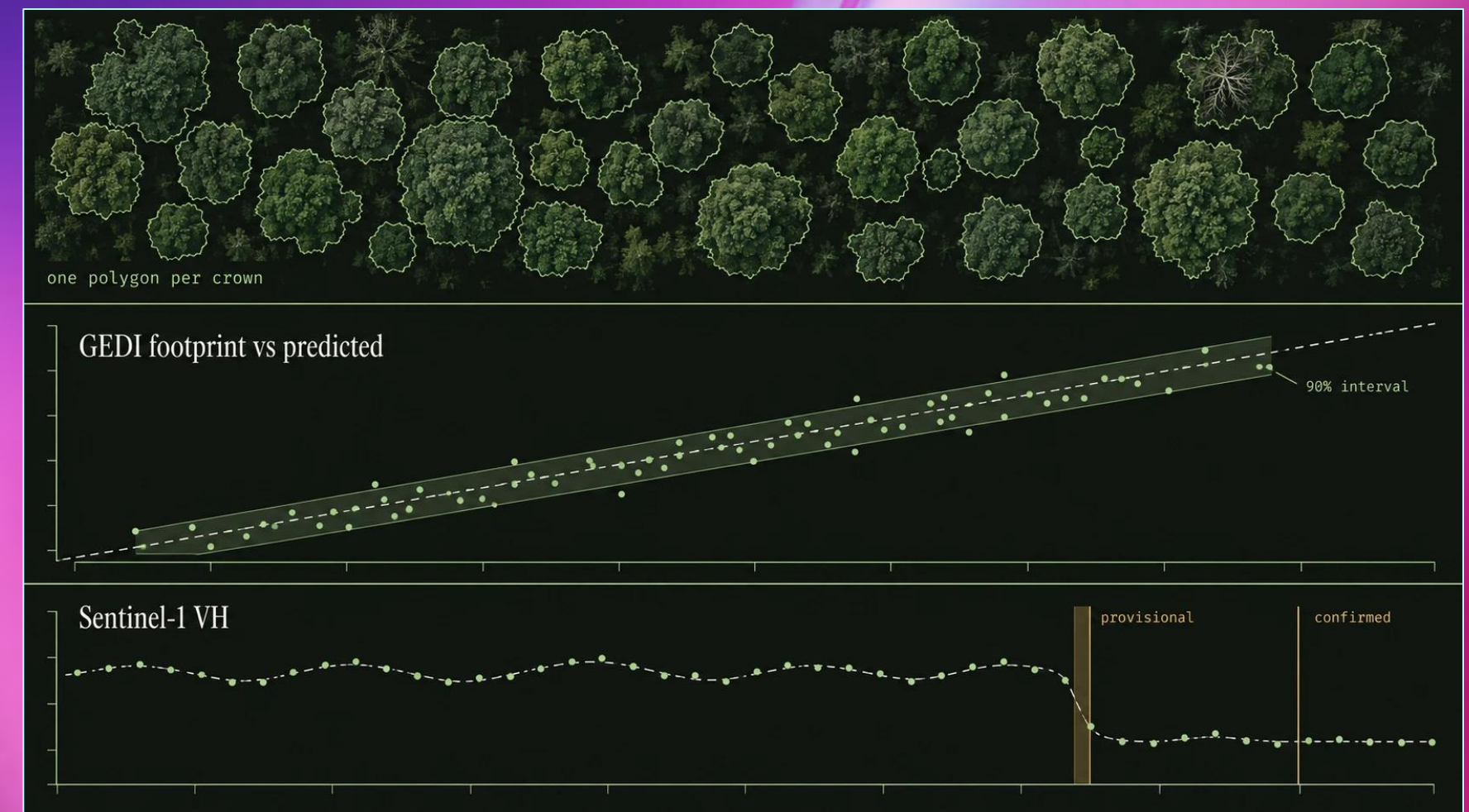
CHANGE / MULTI-YEAR OBSERVATIONS

Harmonic baseline + Bayesian confirmation

A drop must persist before it is called a detection

Below its power threshold the detector reports nothing

$$\text{AGB} = \sum \text{density} \times \text{area} \quad | \quad \text{Aboveground carbon} = \text{AGB} \times \text{CF}$$





Feasibility & Viability

Three gates. Already passing.

01 VERIFIED THE DATA

6 sites audited on 3 independent layers

2 of 5 candidates failed the audit and were rejected

02 EARNED THE MODEL CHOICE

Spatially-blocked folds, never a random split

Neighbouring footprints are one observation, not two

03 MEASURED THE TRADE-OFFS

False positives on primary forest: 0.01–0.02%

Coverage holds on unseen blocks — and one site fails, visibly

41.6% → 0.01%

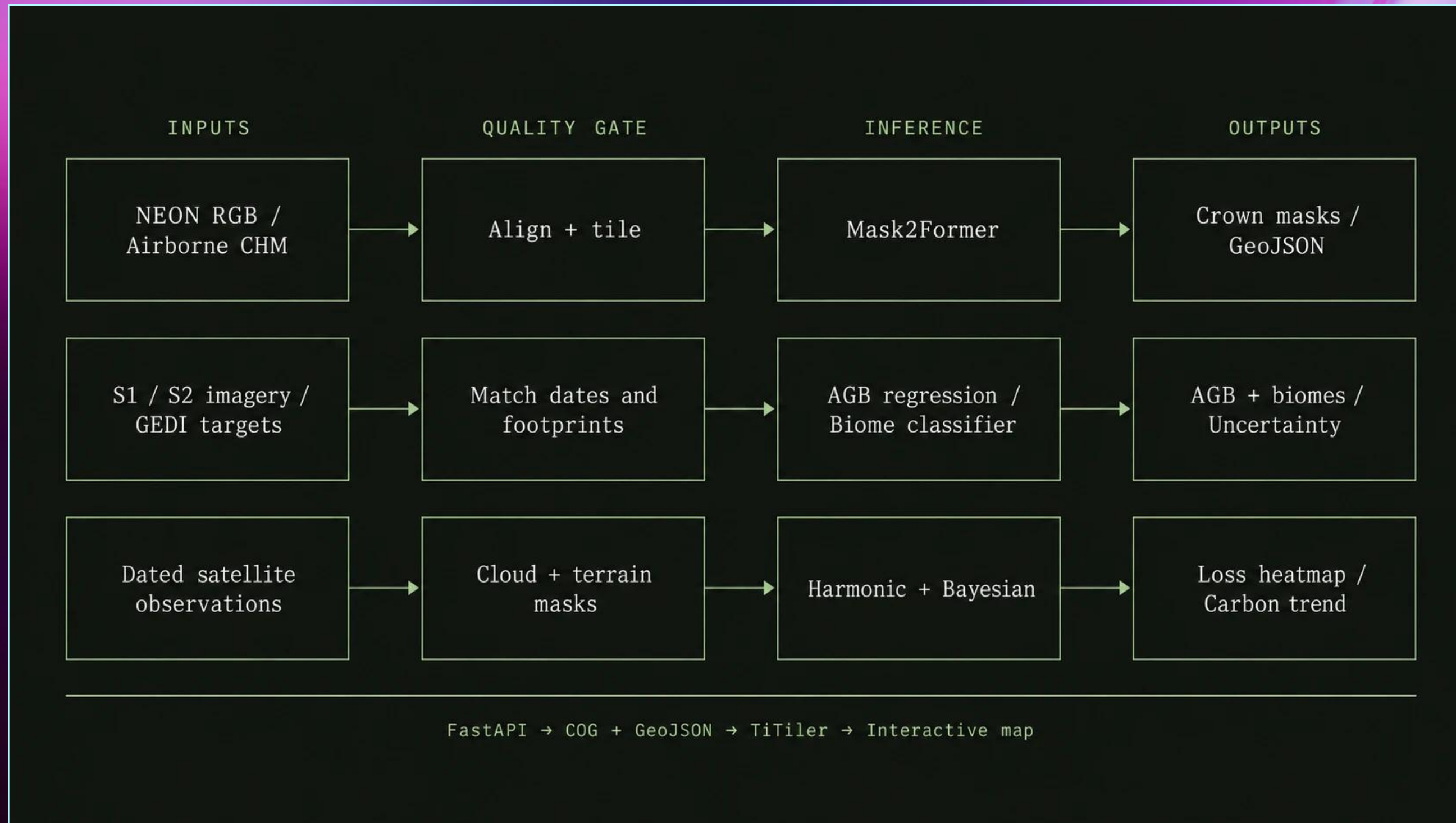
A 365-day window was confirming a national park as cleared.
Persistence is a question about time. The gate is now a span.





Architecture Diagram

One polygon. Three coordinated paths.



Tech stack



DATA

Earth Engine · Planetary Computer · GEDI

PREPARE

Rasterio · GDAL · GeoPandas · odc-stac

MODEL

PyTorch · Mask2Former · split conformal

SERVE

FastAPI · Redis / Celery · Docker

EXPLORE

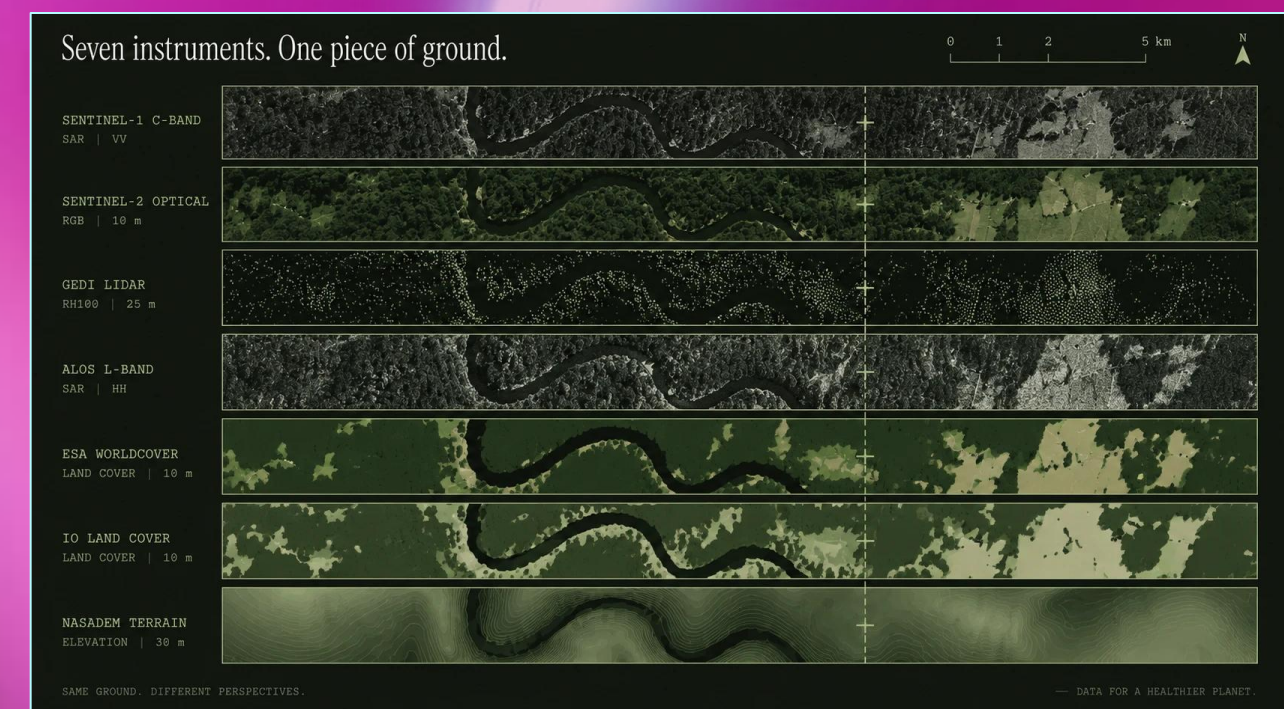
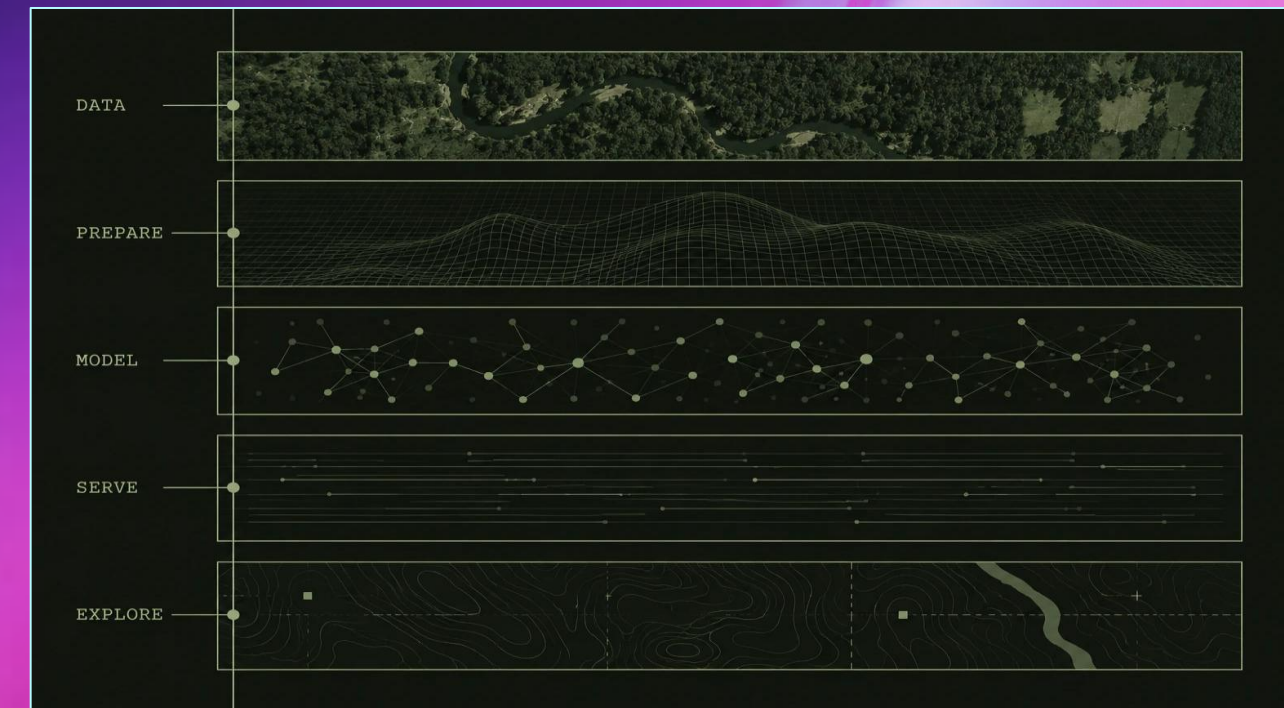
TiTiler · Mapbox GL JS · deck.gl

Windowed COG reads. Cached reports.

Cold report 151–222 s · cached 3–23 ms

7 sources, each free to fail alone

Prescribed stack, with task queues and COG tile serving added for practical inference [1,7,8]



Impact & Benefits

Evidence for every forest decision.

FOREST TEAMS

Prioritise persistent loss alerts

Confirmed and provisional kept apart

RESTORATION TEAMS

Track canopy and biomass trends

Against a measured 2020 baseline

CARBON ANALYSTS

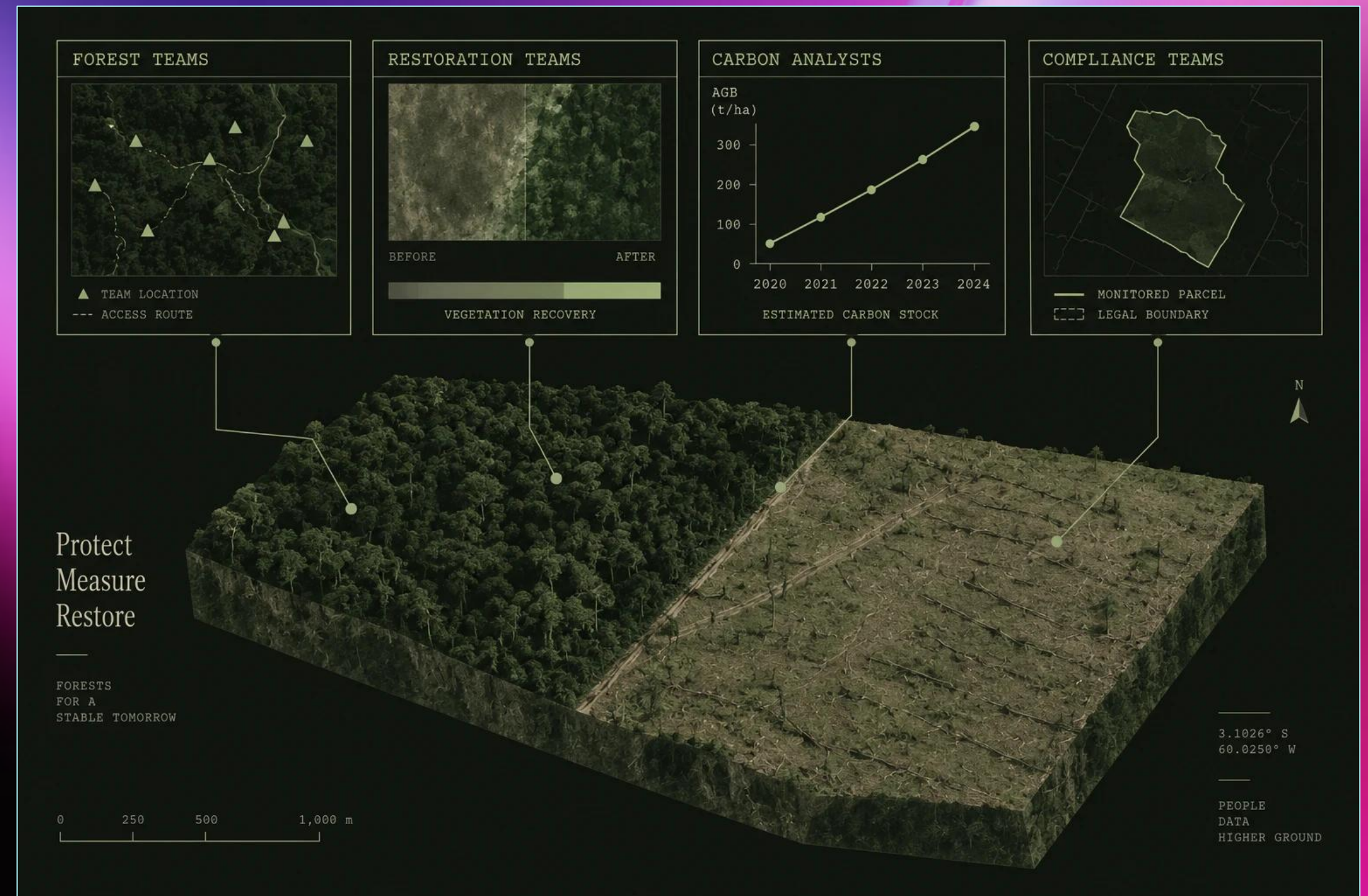
Trace stock change with its interval

Bias-adjusted area, not pixel counts

COMPLIANCE TEAMS

Evidence for EUDR, due Dec 2026

ESA WorldCover 2020 is the cut-off layer





Research & References

01 Official scope + submission rules

[ORION PS-03 · Microsoft Club SIST](#)

02 Satellite observations

[Sentinel-1 processing](#)

[Sentinel-2 resolution](#)

03 Biomass reference targets

[NASA GEDI L4A · footprints + quality](#)

04 Crown-scale data

[NEON 0.1 m RGB](#)

[NEON 1 m CHM](#)

05 Mask2Former + crown evidence

[Cheng et al., 2022](#)

[Nakada et al., 2026 preprint](#)

06 Crown segmentation benchmark

[Detectree2 · Ball et al., 2023](#)

07 Quantile biomass regression

[XGBoost · official quantile objective](#)

08 Time series + raster serving

[Earth Engine CCDC](#)

[TiTiler](#)

09 Biome labels

[RESOLVE ecoregions · Dinerstein et al., 2017](#)

10 Biomass-to-carbon formulation

[IPCC 2006 · Forest Land · Table 4.3](#)